Supplementary Figures

A unified repertoire of indel mutational signatures in 6,975 tumors

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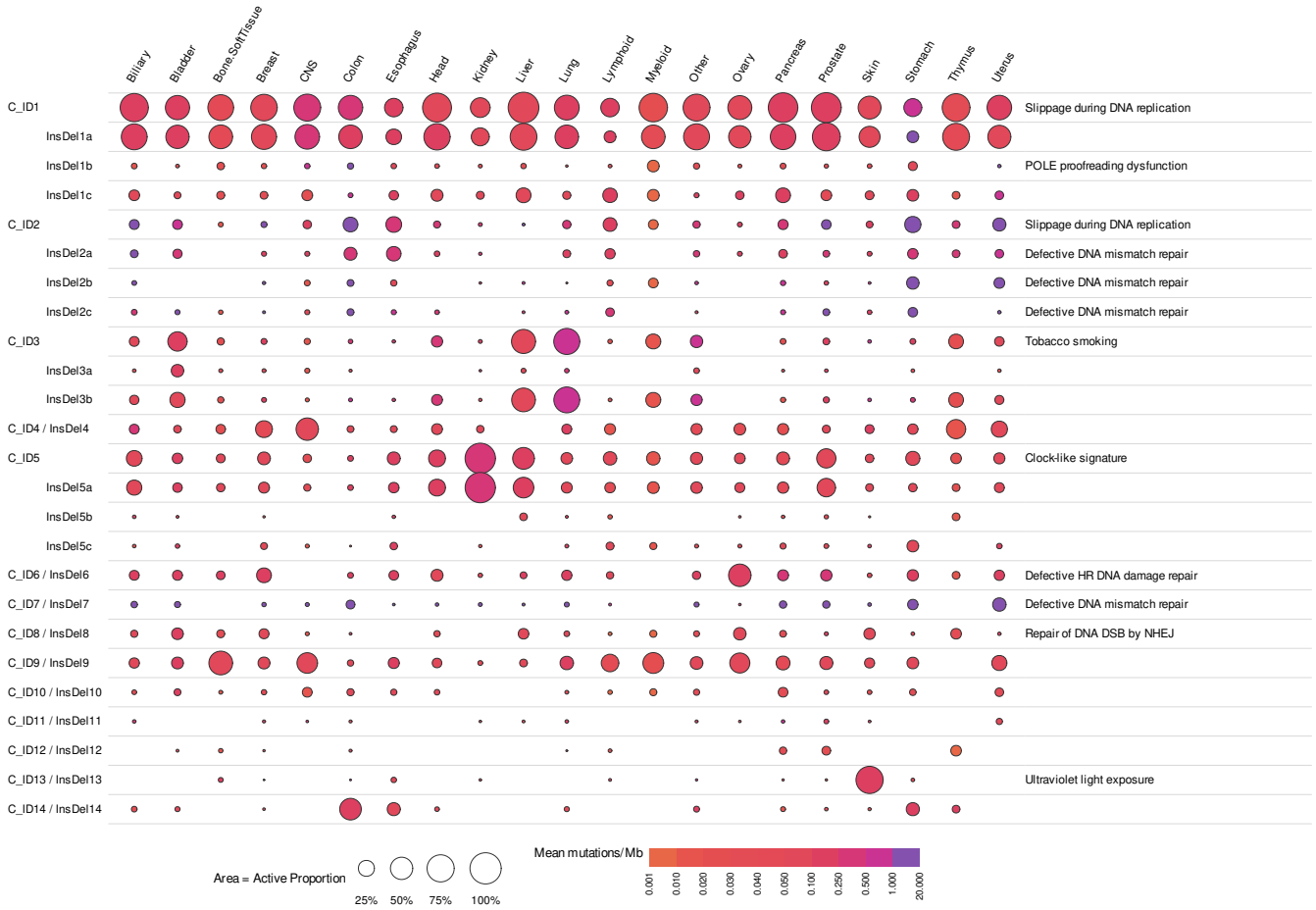


Figure S1. **Activities of 83- and 89-type mutational signatures across cancer types**

The area of each dot represents the proportion of genomes of the given cancer type with activity of the signature. The color indicates the mean number of mutations per megabase attributed to the signature, among the genomes of that cancer type with activity of the signature. Only genomes whose 83-type or 89-type signature assignment reconstructs the spectrum with cosine similarity > 0.90 are included. A row labeled with an 83-type and an 89-type signature (for example C_ID4 / InsDel4) shows a pair with identical activities. The figure continues on the next page.

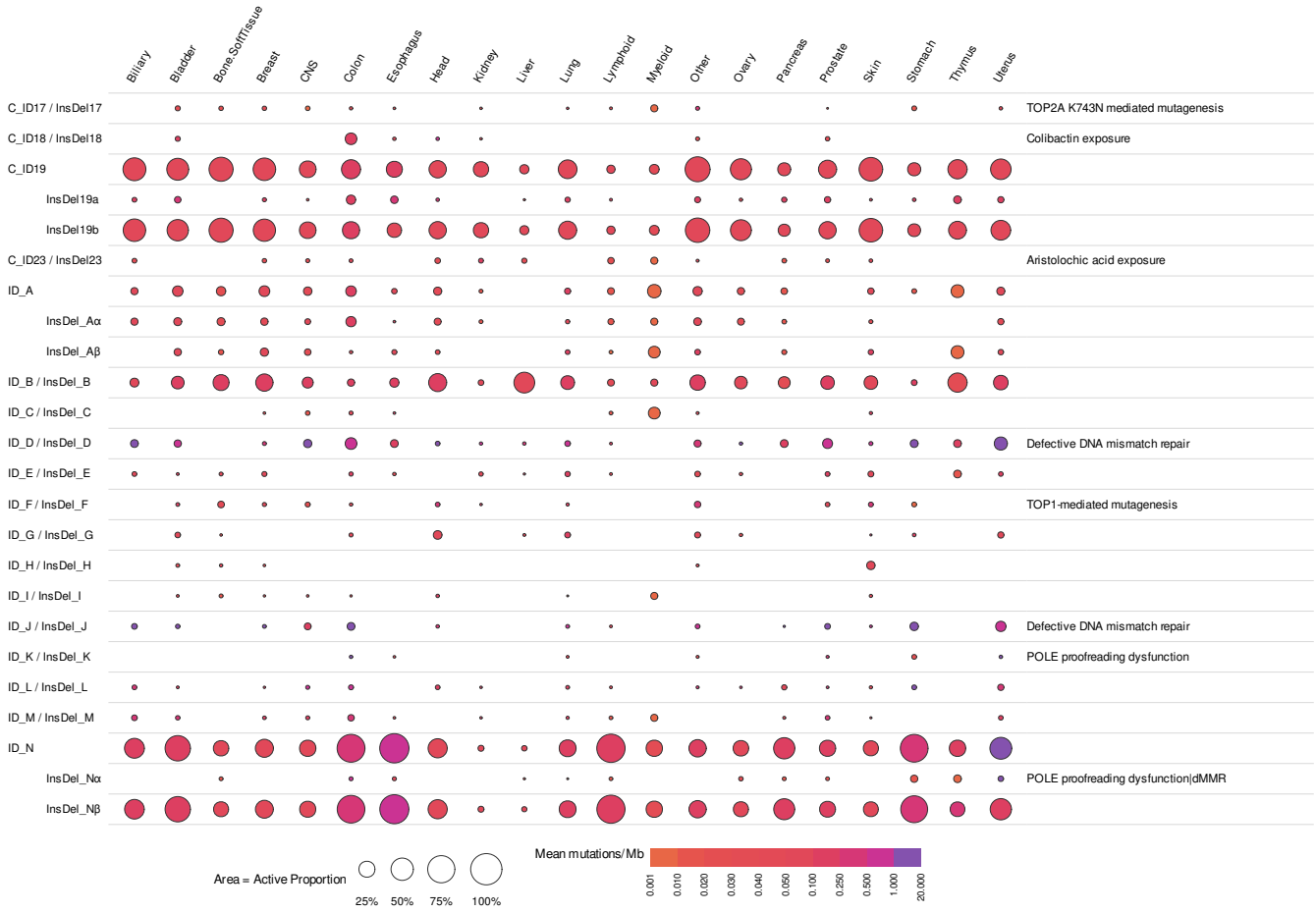


Figure S1 (continued).

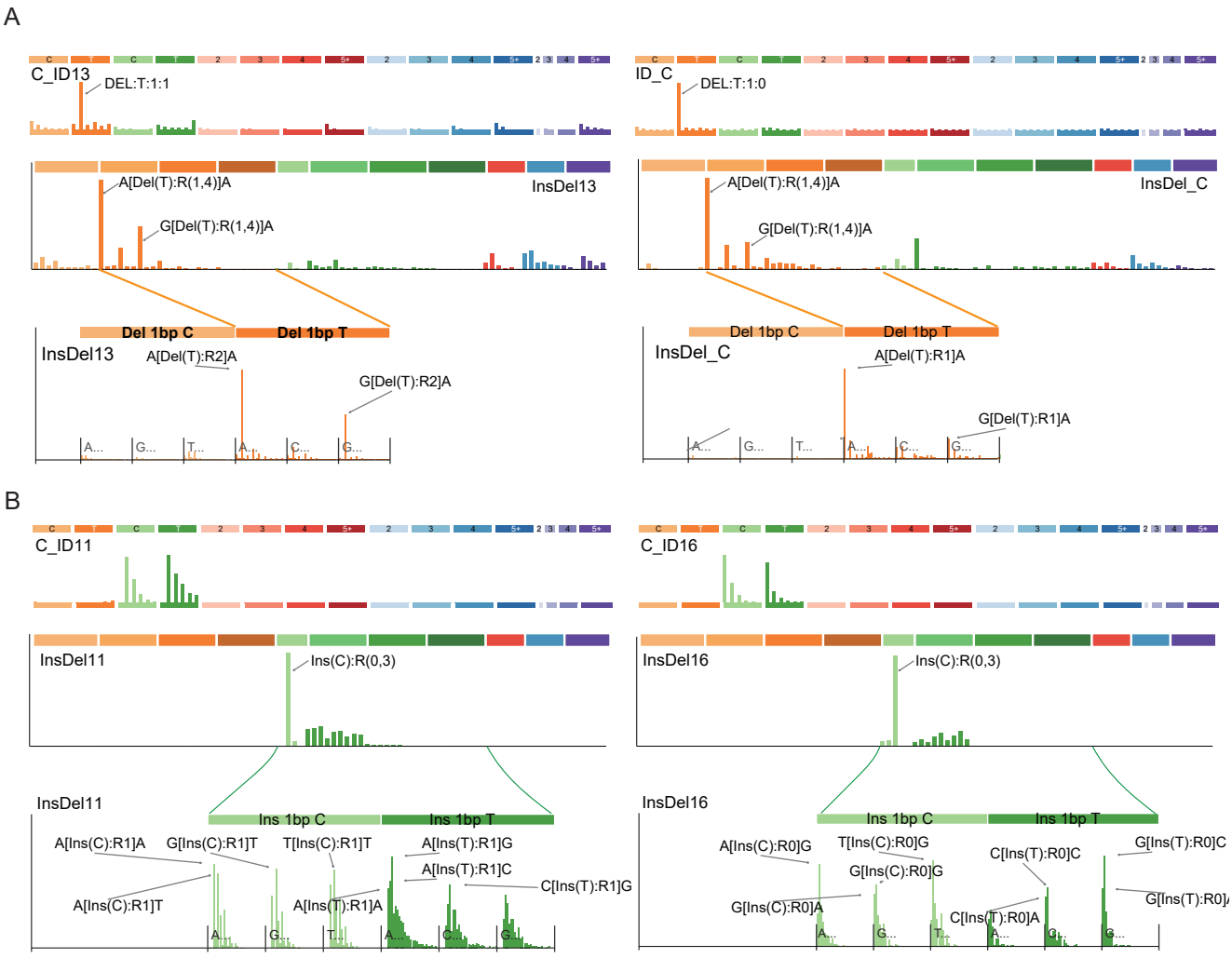


Figure S2. **Comparative mutational signatures across 83-, 89-, and 476-channel classification of C_ID13 vs ID_C, C_ID11 vs C_ID16, and C_ID9 vs ID_A**

(**A**) Divergent sequence contexts for C_ID13 and ID_C. Top to bottom: 83-type, 89-type, and partial 476-type representations (1-bp C and T deletions were displayed). While 89-type profiles exhibit high similarity between the two signatures, the 476-type classification reveals distinct flanking sequence preferences. Specifically, C_ID13 (InsDel13) demonstrates a predilection for ATTA and GTTA motifs, whereas ID_C (InsDel_C) preferentially occurs at ATA and GTA contexts.

(**B**) Divergent sequence contexts for C_ID11 and C_ID16. Top to bottom: 83-type, 89-type, and partial 476-type representations (1-bp C and T insertions were displayed). While 89-type profiles exhibit high similarity between the two signatures, the 476-type classification reveals distinct flanking sequence preferences. Specifically, C_ID11 (InsDel11) exhibits a specific preference for 1-bp T insertions within NTN motifs, while C_ID16 (InsDel16) shows a broader preference for NN contexts. Panel C is on the next page.

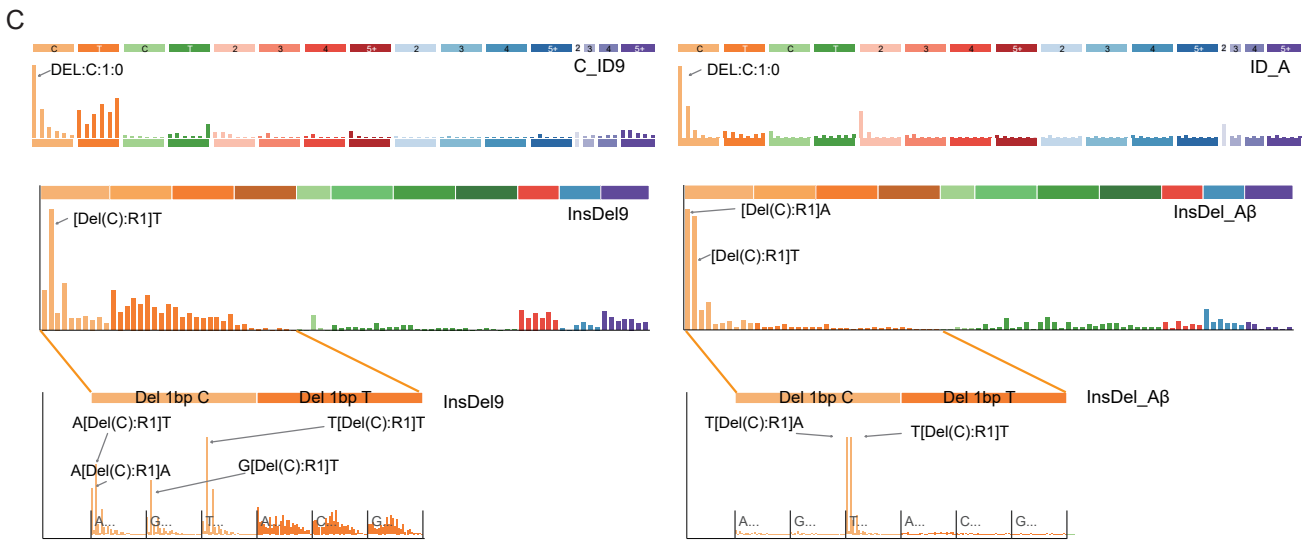


Figure S2 (continued).

(**C**) Divergent sequence contexts for C_ID9/InsDel9 and ID_A/InsDel_Aβ. Top to bottom: 83-type, 89-type, and partial 476-type representations (1-bp C and T deletions were displayed). Compared with C_ID9, ID_A is nearly depleted of 1-bp T deletions. Although the 83-channel profiles show similar patterns of 1-bp C deletions, the 89- and 476-channel classifications reveal distinct preferences for flanking sequences. In particular, InsDel_Aβ preferentially exhibits 1-bp C deletions in TCA and TCT contexts.

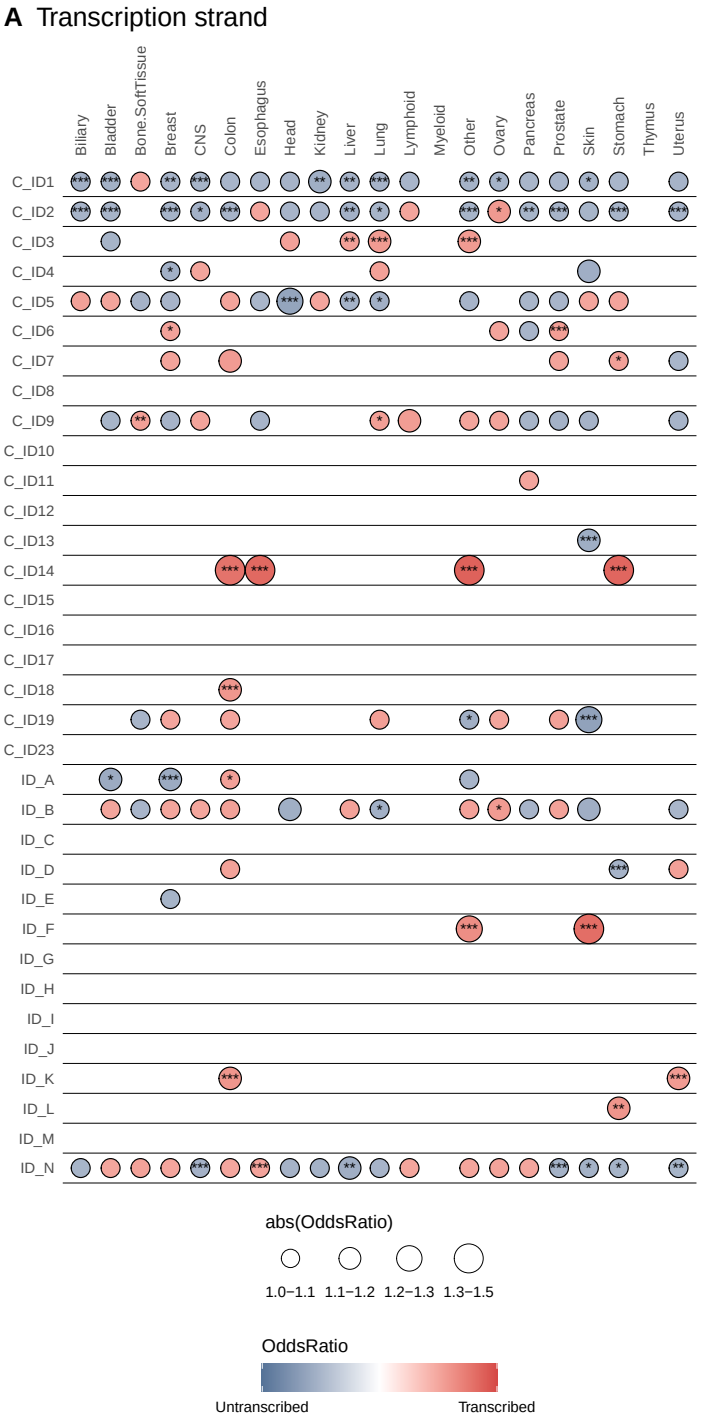


Figure S3. **83-type mutational signature asymmetries in transcription strands, replication strands, and DNA regions across cancer types**

(**A**) Transcription strand asymmetries of 83-type signatures. Each row represents an 83-type signature, and each column represents a cancer type. Red and dark blue circles indicate enrichment on the transcribed and untranscribed strands, respectively. Significance of enrichment is shown as stars (Fisher's exact test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$). Color intensities and circle sizes represent the odds ratio between the ratio of real mutations and the ratio of simulated mutations (see Methods). Panels B and C are on the next two pages.



Figure S3 (continued).

(**B**) Replication strand asymmetries of 83-type signatures. Data are presented as in panel A, with teal and salmon circles indicating enrichment on the leading and lagging strands, respectively.

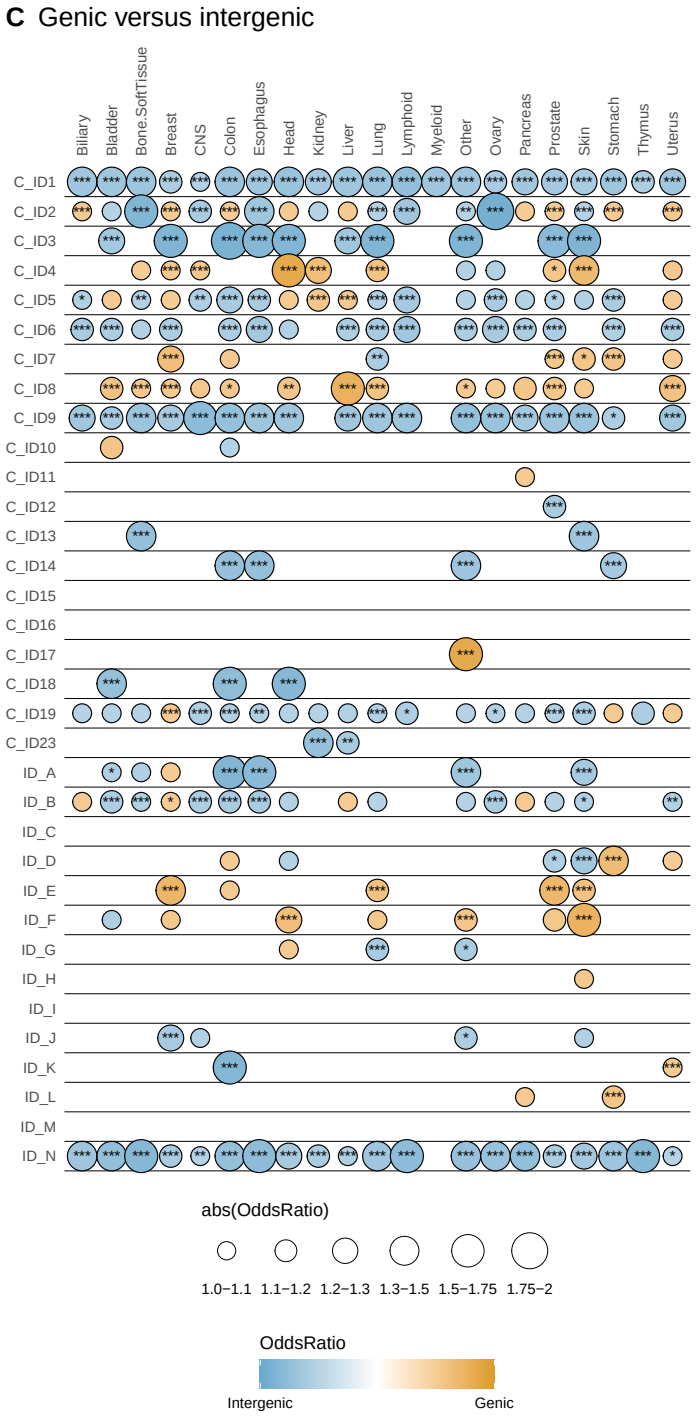


Figure S3 (continued).

(**C**) Enrichment of mutations of 83-type signatures in genic and intergenic regions. Data are presented as in panel A, with orange and light blue circles indicating enrichment in genic and intergenic regions, respectively.



Figure S4. **Interplay between 83-type signatures and replication time**

Relative mutation counts of 83-type signatures per replication-timing decile (y axes) are presented for early (left) to late (right) replication domains. Real mutations are shown as bars, and simulated mutations are shown as red dashed lines. Real and simulated counts are each scaled to their own maximum, so the y axis shows the count relative to the decile with the most mutations. The numbers above each plot are the numbers of cancer types in which the signature increases, decreases, or is flat from early to late replication. The trends are determined separately for each cancer type and relative to the simulated mutations, so they can differ from the pooled real counts shown as bars. Bars sum the cancer types with a trend, that is, those with at least 500 real indels attributed to the signature in which the real or the simulated slope was significant. Green bars indicate signatures that increase from early to late replication in every such cancer type. Yellow bars indicate signatures that decrease in every such cancer type. Purple bars indicate signatures that are flat (not affected by replication timing) in more than half of them. Blue bars indicate all other signatures, whose trends are inconsistent across cancer types.

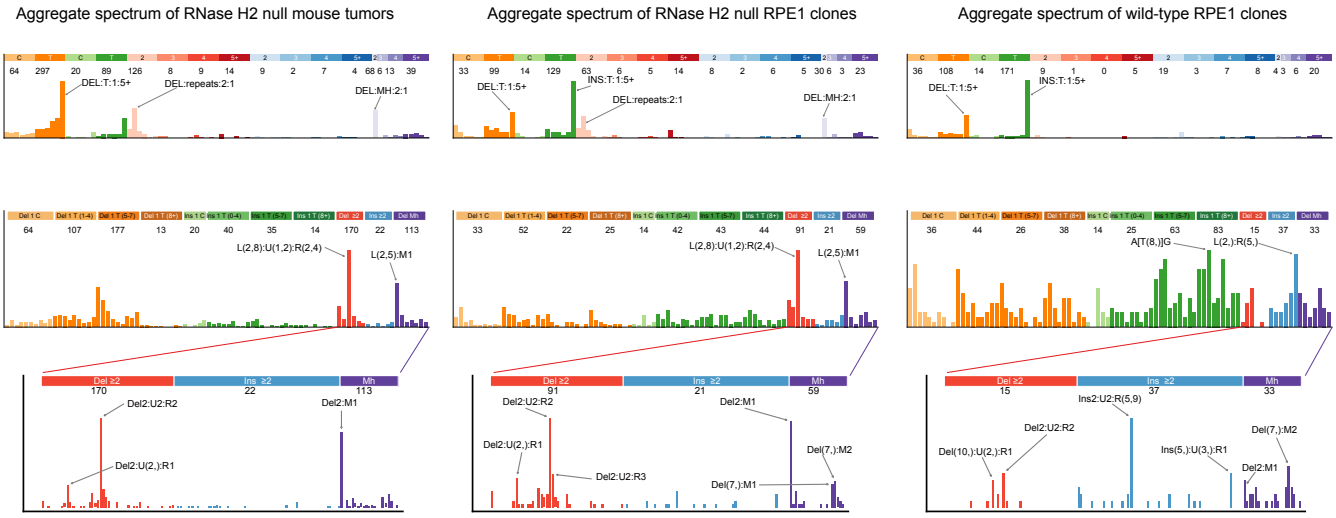


Figure S5. **Mutational spectra of RNase H2-deficient murine tumors and RPE1 cell clones**

From top to bottom, spectra are displayed in 83-type, 89-type, and partial 476-type representations (specifically highlighting ≥ 2-bp deletions, ≥ 2-bp insertions, and deletions with microhomology). Predominant peaks for ID_F/InsDel_F are evident in both RNase H2-deficient tumors and RPE1 clones, characterized by Del:2bp:Repeats:2 and Del:2bp:MH:1 (83-type), L(2,8):U(1,2):R(2,4) and L(2,5):M1 (89-type), and Del2:U2:R2 and Del2:M1 (476-type). Notably, these features are nearly absent in wild-type RPE1 control clones.

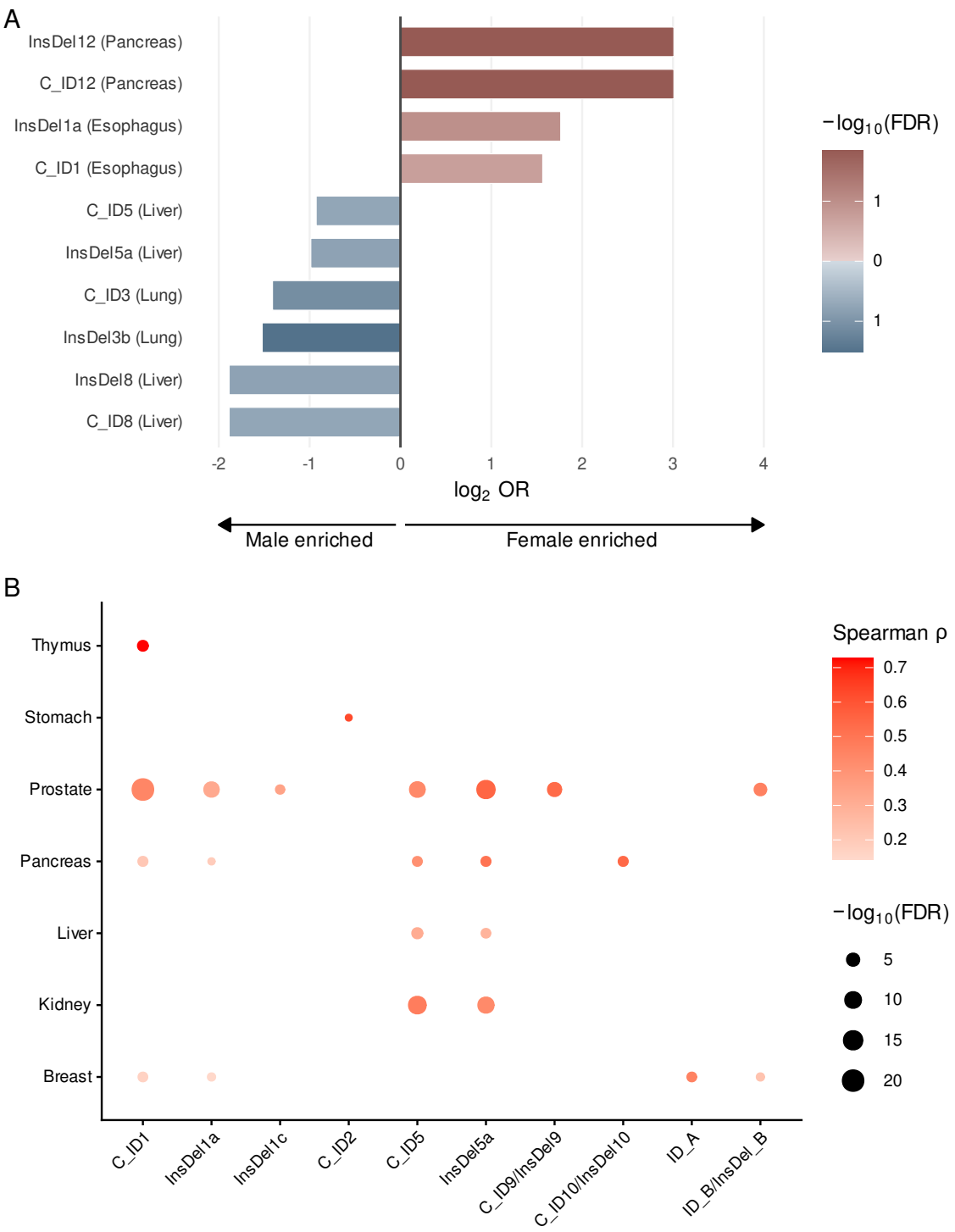


Figure S6. **Demographic associations of indel signatures with sex and age**

(**A**) Sex association of 83-type and 89-type indel signatures. Bar height represents the log□ odds ratio (log□OR), with positive values indicating enrichment in females (red) and negative values indicating enrichment in males (blue). Color shade indicates the significance level (−log□□FDR, where FDR is the Benjamini-Hochberg false discovery rate). Only signatures with FDR < 0.2 are shown.

(**B**) Age association of 83-type and 89-type indel signatures across cancer types. Color shade indicates the Spearman correlation coefficient (ρ), and dot size corresponds to the significance level (−log□□FDR). Only signatures with FDR < 0.05 and $\rho > 0.1$ are shown.